Eberhard Karls Universität Tübingen | Seminar für Sprachwissenschaft (SfS)
Methods I: Logic, WS 2026/27 | Dr. Kyle Richardson
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# Exercise Sheet 2
# Relations & Functions

**Exercise 1:** [4 points] Ever since learning about relations, Alex, our enthusiastic logician, has started wondering whether human relationships are as logical as mathematical ones.

For each of the following relations, mark in the table which properties it satisfies.

(a) Let $R_1$ be a relation on the set of university students, defined as:

*"x and y are friends."*

Assume friendship is mutual and nobody is their own friend.

(b) Let $R_2$ be a relation on the set of human beings, defined as:

*"x is strictly older than y."*

(c) Let $R_3$ be a relation on the set of human beings, defined as:

*"x and y have the same birthday (day and month)."*

(d) Let $R_4$ be a relation on the set of human beings, defined as:

*"x is an ancestor of y."*

Assume nobody is their own ancestor.

| | (a) | (b) | (c) | (d) |
|---|---|---|---|---|
| reflexive | | | | |
| irreflexive | | | | |
| symmetric | | | | |
| asymmetric | | | | |
| anti-symmetric | | | | |
| transitive | | | | |
| intransitive | | | | |
| connected | | | | |

**Exercise 2:** [4 points] Let $X = \{a, b, c, d\}$ and $Y = \{1, 2, 3\}$.

For each of the following, determine whether it is a **function** or only a **relation** between the specified sets. If it is not a function, briefly explain why.

(a) Let $R_1 \subseteq X \times Y$,

$$R_1 = \{\langle a, 1\rangle, \langle b, 2\rangle, \langle c, 3\rangle, \langle d, 1\rangle\}$$

(a) _____

(b) Let $R_2 \subseteq X \times Y$,

$$R_2 = \{\langle a, 1\rangle, \langle b, 2\rangle, \langle c, 2\rangle, \langle d, 3\rangle, \langle a, 3\rangle\}$$

(b) _____

(c) Let $R_3 \subseteq X \times Y$,

$$R_3 = \{\langle a, 1\rangle, \langle b, 2\rangle, \langle c, 3\rangle\}$$

(c) _____

(d) Let $X' = \{a, b, c\}$ and $R_4 \subseteq X' \times Y$,

$$R_4 = \{\langle a, 2\rangle, \langle b, 3\rangle, \langle c, 1\rangle\}$$

(d) _____

For the **functions**, determine whether they are injective, surjective, or bijective. Mark the corresponding properties in the table.

| | (a) | (b) | (c) | (d) |
|---|---|---|---|---|
| injection | | | | |
| surjection | | | | |
| bijection | | | | |

**Exercise 3:** [4 points] Let $X = \{a, b, c\}$ and $Y = \{1, 2, 3, 4\}$, and let $R \subseteq X \times Y$ be defined as follows:

$$R = \{\langle a, 1\rangle, \langle a, 3\rangle, \langle b, 2\rangle, \langle c, 1\rangle, \langle c, 4\rangle\}.$$

(a) [1 point] Determine the domain and range of $R$.

(b) [1 point] Determine the converse $R^{-1}$.

(c) [1 point] Determine the complement $(X \times Y) \setminus R$.

(d) [1 point] Is $R$ a function from $X$ to $Y$? Justify your answer.

**Exercise 4:** [4 points] Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = 3x - 5.$$

(a) [2 points] Prove that $f$ is injective.

(b) [2 points] Prove that $f$ is surjective.